



Various types of Capacitors

In this unit student should be able to:

- Explain capacitors as Charge storing Devices.
- Identify types of capacitors used in different field.
- Identify factors affecting the capacitance of a parallel plate capacitor and use equations
$$\epsilon_r = C/C_0; \quad C = \epsilon_0 \epsilon_r A/d$$
- Calculate combined capacitance of capacitors in series and in parallel.
- Demonstrate charging and discharging of a capacitor through a resistance.
- Prove that energy stored in capacitor is
$$W = \frac{1}{2} QV \text{ and hence } E = \frac{1}{2} CV^2$$

9.1.1 Capacitor

A capacitor is an electrical device which is used to store electrical energy in the same way as the bucket is used for storing water or a tank for storing gas. Each of these devices has fixed capacity which does not depend on the quantity to be stored.

Capacitors are electronic components designed to store electric charge. Figure 9.1 shows the schematic diagram of parallel plate capacitors. They consist of two conductive plates separated by an insulating material called a dielectric. When a voltage is applied across the plates, one plate accumulates a positive charge while the other accumulates an equal and opposite negative charge, resulting in the storage of electric charge in the capacitor.

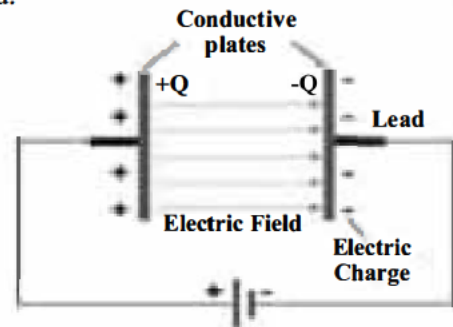


Fig: 9.1

A Parallel Plate Capacitor.

The ability of a capacitor to store charge is determined by its capacitance (C), which is measured in farads (F). The capacitance depends on the physical characteristics of the capacitor, such as the area of the plates, the distance between them, and the properties of the dielectric material.

The quantity of charge Q on a capacitor is directly proportional to the potential difference V between the plates; that is, $Q \propto V$.

$$Q = CV \quad (9.1)$$

Where:

Q = Electric charge stored in the capacitor (measured in coulombs, C)

C = Capacitance of the capacitor (measured in farads, F)

V = Voltage applied across the capacitor (measured in volts, V)

The proportionality constant depends on the shape and separation of Plates.

The capacitance of a capacitor is defined as

“The amount of charge required to create a unit potential difference between Parallel plates.”

From equation (9.1), the capacitance C of a capacitor can be measured as

$$C = \frac{Q}{V}$$

Hence, capacitance is the charge stored per unit potential difference.

DO YOU KNOW?

Capacitors are very useful to reduce voltage fluctuations in electronic power supplies, to transmit signals, to detect electro-magnetic oscillations at radio frequencies and to other important electronic circuits.

DO YOU KNOW?



Capacitor also used as sensor

9.1.2 Types of Capacitor

Capacitors play a very important role in various electrical circuits, especially radio circuit. Here we classify some common types of capacitors.

Variable Capacitors:

This type of capacitor is used to control the capacitance continuously for tuning transmitters and receivers signals.



Variable capacitor

Film Capacitors:

Film Capacitors consisting of a relatively large family of capacitors with the difference being in their dielectric properties. They are available in almost any value and voltages as high as 1500 volts. Film capacitors are used in power electronics devices, phase shifters, X-ray flashes and pulsed lasers.



Film capacitor

Ceramic capacitors:

These capacitors are used in high-frequency circuits ranging from audio to radio frequency. These capacitors are also called disc capacitors.



Ceramic capacitor

Electrolytic Capacitors:

These capacitors are generally used when very large capacitance values are required. The majority of electrolytic capacitors are polarized. Electrolytic Capacitors are generally used in DC power supply circuits due to their large capacitance's and small size which help to reduce the ripple voltage.



Electrolytic capacitor

9.1.3 Capacitance of a parallel plate Capacitor

It consists of two parallel metallic plates of equal area A which are separated by a distance d as shown in **figure 9.2**. Both parallel plates carry charges of magnitude $+Q$ and $-Q$ respectively. The surface charge density (σ) is defined as the charge stored (Q) per unit area (A).

$$\sigma = Q / A$$

Therefore, the electric field in this case will be

$$E = \frac{Q}{A\epsilon_0} = \frac{\sigma}{\epsilon_0} \quad \dots \dots (9.2)$$

As the field between the plates is uniform, therefore, the magnitude of the potential difference between the plates equals Ed . i.e., $V=Ed$, Substituting the value of V from equation (9.1), we get,

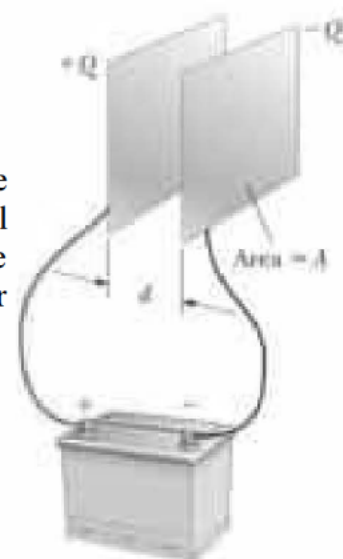


Fig: 9.2

A parallel-plate capacitor consists of two parallel conducting plates, each of area A , separated by a distance d .

$$E = \frac{V}{d} = \frac{Q}{Cd} \quad \dots \dots (9.3)$$

By comparing the equations (9.2) and (9.3), we get relation for capacitance as given below

$$C = \frac{\epsilon_0 A}{d} \quad \dots \dots (9.4)$$

Equation (9.4) shows that the capacitance of a parallel-plate capacitor is proportional to the area of plates and inversely proportional to the plate separation.

Effects of dielectric materials:

Experimentally it is found that capacitance of a capacitor can be increased by filling the space between the plates with dielectrics. Dielectrics are electrically insulating materials that increase the ability of storing charges on the plates of the capacitor. Polythene and waxed paper are the examples of dielectrics.

Microscopically, the molecules of the dielectric materials become polarized by the charged plates of the capacitor as shown in figure 9.3. This polarization is responsible to increase the capacitance of the capacitor. When the dielectric is inserted between the plates of a capacitor, then permittivity of such capacitor will increase by the amount of $\epsilon_0 \epsilon_r$, where ϵ_r is the relative permittivity of a substance and it is called the dielectric constant.

Therefore, the capacitance of such capacitor is given as

$$C_o = \frac{\epsilon_0 \epsilon_r A}{d}$$

$$C_o = \epsilon_r \left(\frac{\epsilon_0 A}{d} \right) = \epsilon_r C$$

The above relation shows that the charge storing capacity of a capacitor is enhanced by the dielectric which permits it to store ϵ_r times more charge for the same potential difference.

$$\epsilon_r = \frac{C_o}{C} \quad \dots \dots (9.5)$$

The relative permittivity ϵ_r of the material is the ratio of the capacitance of a capacitor with a given materials filling the space between the plates to the capacitance of the same capacitor when space is evacuated. Typical values of ϵ_r are presented in table 9.1.

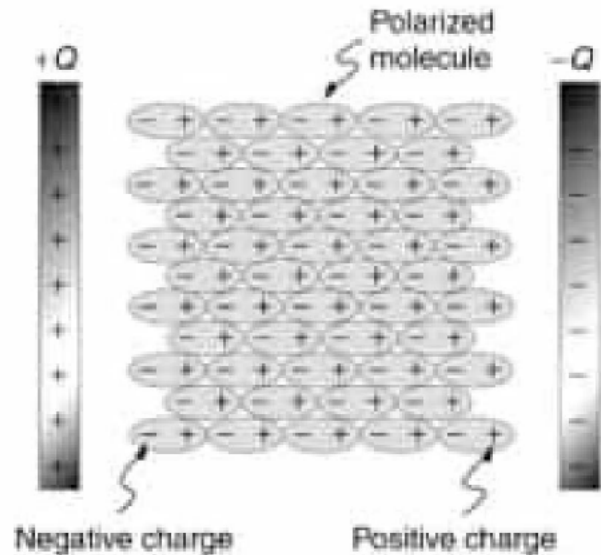


Fig: 9.3
A capacitor filled with dielectric material.

Table No. 9.1: Dielectric Constant (ϵ_r)

S/No.	Insulating material	Dielectric constant
1	Vacuum	1
2	Air	1.0006
3	Teflon	2
4	Mineral Oil	2.2
5	Polyethylene	2.3
6	Waxed paper	2.5
7	Epoxy	3.3
8	PVC	3.7
9	Nylon	4.1
10	Bakelite	5
11	Mica	7
12	Water	81

Worked Example 9.1

A parallel plate capacitor consists of two plates with an area of 0.01 m^2 each, separated by a distance of 0.002 m . The capacitor is filled with a dielectric material having a relative permittivity (ϵ_r) of 4. Calculate the capacitance of this capacitor.

Solution:

Step 1: Formula:

$$C = \epsilon_0 \epsilon_r A / d$$

where:

C is the capacitance,

ϵ_0 is the vacuum permittivity constant (approximately $8.854 \times 10^{-12} \text{ F/m}$),

ϵ_r is the relative permittivity (given as 4),

A is the area of one plate (0.01 m^2),

d is the separation distance between the plates (0.002 m).

Step 2:

Now, substitute the given values into the formula:

$$C = (8.854 \times 10^{-12} \text{ F/m}) \times 4 \times (0.01 \text{ m}^2) / (0.002 \text{ m})$$

$$C \approx 8.854 \times 10^{-12} \times 4 \times (0.01 / 0.002) \text{ F}$$

$$C \approx 3.5416 \times 10^{-11} \text{ F}$$

So, the capacitance of the parallel plate capacitor is approximately $3.5416 \times 10^{-11} \text{ Farads (F)}$.

Self-Assessment Questions:

1. The capacitance of a capacitor formed by two parallel metal plates each 200 cm^2 in area separated by a dielectric 4 mm thick is 0.0004 microfarads. A potential difference of 20,000 V is applied. Calculate the total charge on the plates?

9.1.4 Combinations of Capacitors

Two or more capacitors often are combined in electric circuits. We can calculate the equivalent capacitance of certain combinations with the help of a circuit diagram. The circuit symbols for capacitors, batteries, and switches are shown in figure 9.4.

Parallel Combination of capacitors

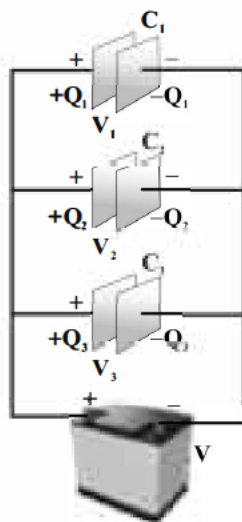
In a parallel combination of capacitors, two or more capacitors are connected side by side with their positive terminals connected together and their negative terminals connected together. This arrangement allows the capacitors to share the same voltage across their terminals while increasing the total capacitance of the combination.

When capacitors are connected in parallel, the total capacitance (C_{total}) of the combination is the sum of the individual capacitances (C_1 , C_2 , C_3 , and so on) of the capacitors connected:

$$C_{\text{total}} = C_1 + C_2 + C_3 + \dots + C_n$$

Consider three capacitors are connected in parallel combination as shown figure 9.5 and all three diagrams are equivalent. we know that potential difference across each capacitors connected in parallel are the same and equal to the potential difference applied across the combination. i.e.,

$$V = V_1 = V_2 = V_3$$



(a) a pictorial representation of three capacitors connected in parallel to a battery

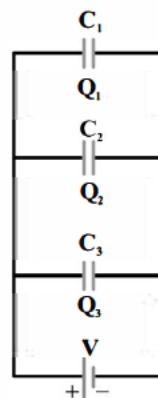
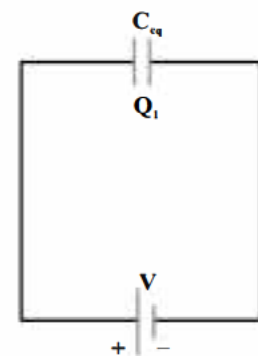


Fig: 9.5

(b) circuit diagram showing the three capacitors connected in parallel to a battery



(c) circuit diagram showing the equivalent capacitance of the capacitors connected in parallel.

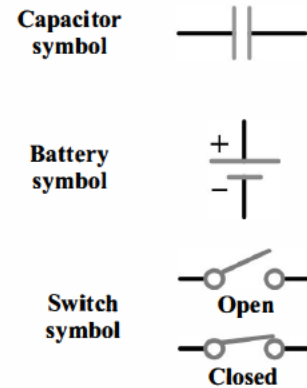


Fig: 9.4

Circuit symbols for capacitors, batteries, and switches.

In general, when a potential difference V is applied across several capacitors connected in parallel, that potential difference V is same across each capacitor. The total charge Q stored on the equivalent capacitor is the sum of the charges stored on all the capacitors.

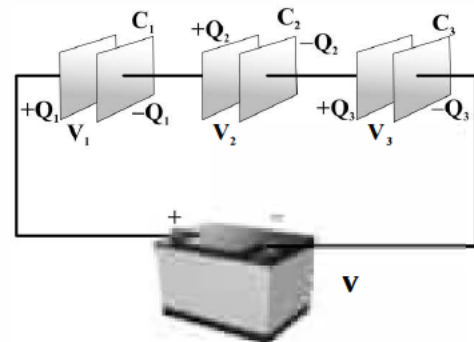
To find an expression for equivalent capacitance C_{eq} , use the eq. (9.1) to find the charge on each actual capacitor and equivalent capacitor.

$$Q_1 = C_1 V, Q_2 = C_2 V, Q_3 = C_3 V \text{ and } Q = C_{eq} V$$

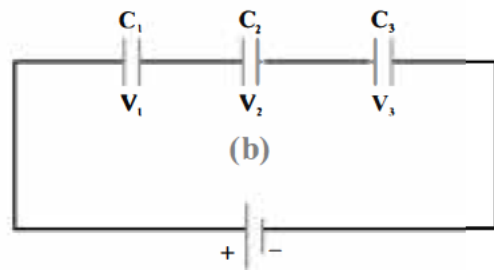
The total charge on the parallel combination is given as

$$\begin{aligned} Q &= Q_1 + Q_2 + Q_3 \\ C_{eq} V &= C_1 V + C_2 V + C_3 V \\ C_{eq} &= C_1 + C_2 + C_3 \quad \dots (9.6) \end{aligned}$$

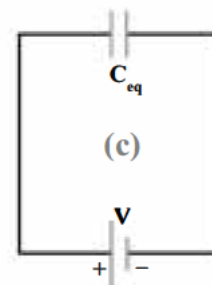
Therefore, the equivalent capacitance of a parallel combination of capacitors is the algebraic sum of the individual capacitances and has greater value than any of the individual capacitances. This combination is used in the circuit where the high capacitance is required because equivalent capacitor store more energy.



(a)



(b)



(c)

Fig: 9.6

(a) a pictorial representation of two capacitors connected in series to a battery

(b) circuit diagram showing the two capacitors connected in series to a battery

(c) circuit diagram showing the equivalent capacitance of the capacitors connected in series

Series Combination of capacitors

In a series combination of capacitors, two or more capacitors are connected one after the other in a single line, with the positive terminal of one capacitor connected to the negative terminal of the next capacitor. This arrangement allows the capacitors to share the same charge while effectively reducing the total capacitance of the combination compared to individual capacitors.

Three capacitors are connected in series combination as shown Figure 9.6 (a, b, c) and all three diagrams are equivalent. In series means that the capacitors are connected by a wire and there is no other way of flowing of charge. When a potential V is applied across them, the same amount Q of charge will appear on each of the capacitor. The battery directly produces charges on only the two plates to which it is connected. Charges that are produced on the other plates are due to the simple impact of electric field induction. Due to this the charges can be distributed over all other plates connected in the circuit. Therefore, the charges on capacitors connected in series are the same.

In general, the total potential difference across any number of capacitors connected in series is the sum of the potential differences across the individual capacitors.

$$V = V_1 + V_2 + V_3$$

To find an expression for equivalent capacitance C_{eq} , use the eq. (9.1) to find the potential difference across each actual capacitor and equivalent capacitor.

$$V_1 = \frac{Q}{C_1}, V_2 = \frac{Q}{C_2}, V_3 = \frac{Q}{C_3} \text{ and } V = \frac{Q}{C_{eq}}, \text{ therefore,}$$

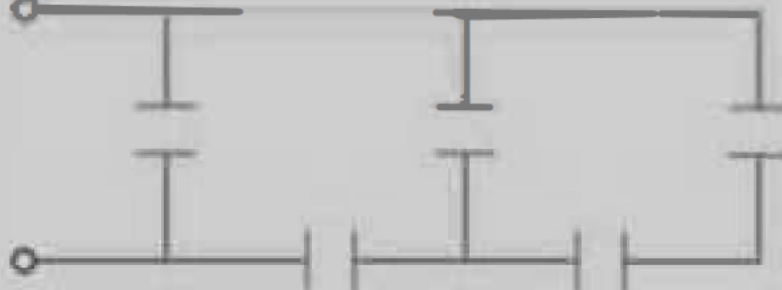
we have

$$\begin{aligned} \frac{Q}{C_{eq}} &= \frac{Q}{C_1} + \frac{Q}{C_2} + \frac{Q}{C_3} \\ \frac{1}{C_{eq}} &= \frac{1}{C_1} + \frac{1}{C_2} + \frac{1}{C_3} \quad \dots \dots (9.7) \end{aligned}$$

This expression shows that the reciprocal of the equivalent capacitance is the algebraic sum of the reciprocal of the individual capacitance. Thus, the equivalent capacitance of a series combination is always less than any individual capacitance.

Self-Assessment Questions:

- Find the equivalent capacitance of the circuit shown in figure and all capacitors have the same capacitance of $15\mu F$.



- Two capacitors X and Y are connected in series across a 100 V supply and it is observed that the potential difference across them are 60 V and 40 V respectively. A capacitor of $2\mu F$ capacitance is now connected in parallel with X and the potential difference across Y rises to 90 volts. Calculate the capacitance of X and Y.

9.2.1 Charging and discharging of a capacitor through a resistor

Consider a circuit having a capacitance C and a resistance R which are connected in series with a battery of potential difference V through a switch as shown in the **figure 9.7**. This circuit is called RC circuit.

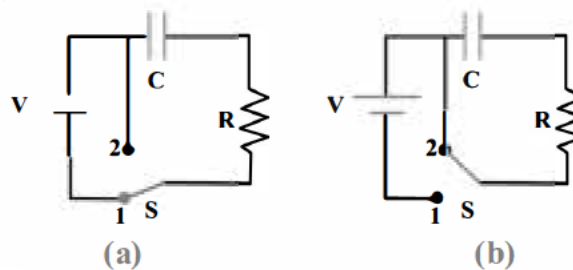


Fig: 9.7 Resistor Capacitor (RC) circuit.

When the switch is at position 1 as shown in Fig. 9.7(a), the capacitor begins to store charge. If at any time during charging, the charge Q flowing through the circuit and Q is the charge on the capacitor, then sum of potential difference between the plates of the capacitor and across the resistor is equal to the potential difference supply from the battery. As the current stops flowing when the capacitor is fully charged, When $Q = Q_0$ (the maximum value of the charge on the capacitor), then

$$Q_0 = C V$$

Experiments shows that the charging process of a capacitor exhibits the exponential behaviour therefore we can write its equation as

$$Q = Q_0(1 - e^{-t/R}) \quad \dots \dots (9.8)$$

where Q_0 represents the final charge on the capacitor that accumulates. Where the quantity $\tau = RC$ is called the time constant or the capacitive time constant of the circuit and it has dimensions of time. Further,

if $RC \ll 1$, Q will attain its final value rapidly

if $RC \gg 1$, it will do so slowly.

Thus, RC determines the rate at which the capacitor charges (or discharges) itself through a resistance.

If $\tau = t$, then from equation (9.8)

$$Q = Q_0(1 - e^{-1}) = Q_0\left(1 - \frac{1}{e}\right)$$

$$Q = Q_0\left(1 - \frac{1}{2.718}\right) = Q_0(1 - 0.368)$$

$$Q = 0.632Q_0 \dots \dots (9.9)$$

Therefore the time constant is the duration of time for the capacitor in which 63.2% of its maximum value charge is deposited on the plates during the charging of the capacitor. The equation (9.8) shows that charge builds up exponentially during the charging process (See Fig. 9.8 a).

When the switch is moved to position 2, for the circuit which shows the discharging of the charged capacitor (see Fig. 9.8(b)). The battery is now out of the circuit and the capacitor will discharge itself through resistor R . This discharging process of the capacitor follows the following equation.

$$Q = Q_0 e^{-t/\tau} \quad (9.10)$$

where Q_0 represents the initial charge on the capacitor at the beginning of the discharge, i.e., at $t=0$. The above expression shows that the charge

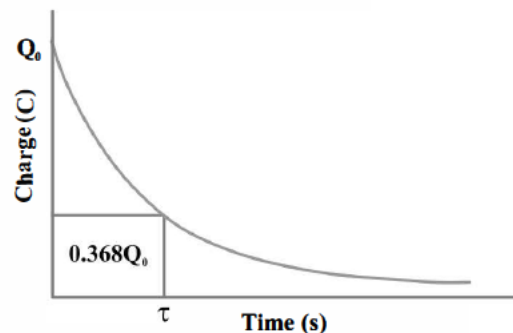
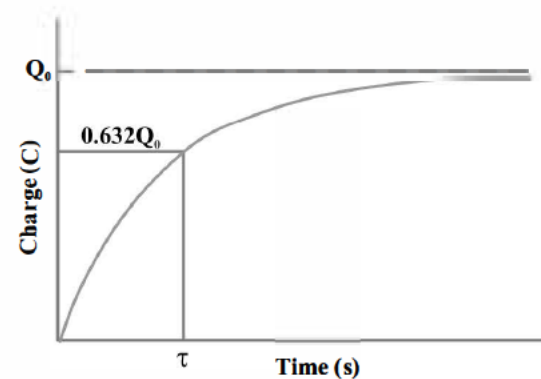


Fig: 9.8
Capacitor (a) charging and (b) discharging verses time.

decays exponentially when the capacitor discharges, and that it takes an infinite amount of time to fully discharge.

If $\tau = RC = t$, the equation (9.9) becomes

$$Q = Q_0 e^{-1} = \frac{Q_0}{e} = \frac{Q_0}{2.718}$$

$$Q = 0.368 Q_0 \dots\dots(9.11)$$

Time constant of a RC circuit is thus also the time during which the charge on the capacitor falls from its maximum value to 0.368 of its maximum value.

9.2.2 Energy Stored in a Capacitor

Charging of a capacitor always involves some expenditure of energy by the charging agency. This energy is stored up in the electrostatic field set up in the dielectric medium. On discharging the capacitor, the field collapses and the stored energy is released. Many of those who work with electronic equipment have at some time verified that a capacitor can store energy. If the plates of a charged capacitor are connected by a conductor such as a wire, charge moves between each plate and its connecting wire until the capacitor is uncharged.

The discharge can often be observed as a visible spark. If you accidentally touch the opposite plates of a charged capacitor, your fingers act as a pathway for discharge and the result is an electric shock. The degree of shock you receive depends on the capacitance and the voltage applied to the capacitor. Such a shock could be dangerous if high voltages are present as in the power supply of a home theater system. Because the charges can be stored in a capacitor even when the system is turned off, unplugging the system does not make it safe to open the case and touch the components inside.

Let us consider a capacitor connected to source of potential difference V . Initially, when the capacitor is uncharged, the potential difference between the plates is zero. Finally when charge $+Q$ and $-Q$ deposited on the plates, the potential difference between the plates becomes V . The average voltage on the capacitor during the charging process is $V/2$. See chapter 08 section 8.5.2, The equation (8.7) can be modified for this case

$$\frac{V}{2} = \frac{U}{Q}$$

$$U = \frac{1}{2} QV \quad \therefore Q = CV$$

DO YOU KNOW?

The flash which comes from the camera when we take photographs is due to the energy released from the capacitor.



DO YOU KNOW?

During cardiac arrest, heart defibrillator is used to give a sudden surge of a large amount of electrical energy to the patient's chest to retrieve the normal heart function.



$$U = \frac{1}{2} CV^2 \quad \dots \dots (9.12)$$

For a given capacitance, the stored energy increases as the charge and the potential difference increase. In practice, there is a limit to the maximum energy (or charge) that can be stored because, at a sufficiently large value of V , discharge ultimately occurs between the plates. For this reason, capacitors are usually labeled with a maximum operating voltage.

Self-Assessment Questions:

1. A $2200 \mu\text{F}$ capacitor is charged to a potential difference of 9.0 V then discharged through a $100 \text{ k}\Omega$ resistor. Calculate (a) the initial charge stored by the capacitor, (b) the time constant of the circuit, (c) calculate the potential difference after a time of 300 s equal to the time constant.
2. An air-capacitor of capacitance $0.005 \mu\text{F}$ is connected to a direct voltage of 500 V , is disconnected and then immersed in oil with a relative permittivity of 2.5 . Find the energy stored in the capacitor before and after immersion.



- A capacitor consists of two isolated plates with charges $+q$ and $-q$. Its capacitance C is defined from $Q = CV$
- where V is the potential difference between the plates. The capacitance depends only on the geometry of the conductors and not on an external source of charge or potential difference. The SI unit of capacitance is coulombs per volt, or the farad (F): $1 \text{ F} = 1 \text{ C/V}$.
- A parallel-plate capacitor with flat parallel plates of area A and spacing d has capacitance $C = \frac{\epsilon_0 A}{d}$
- If the space between the plates of a capacitor is completely filled with a dielectric material, the capacitance C is increased by a factor ϵ_r , called the dielectric constant, which is characteristic of the material. In a region that is completely filled by a dielectric, all electrostatic equations containing ϵ_0 must be modified by replacing ϵ_0 with $\epsilon_0 \epsilon_r$.
- The effects of adding a dielectric can be understood physically in terms of the action of an electric field on the permanent or induced electric dipoles in the dielectric slab. The result is the formation of induced charges on the surfaces of the dielectric, which results in a weakening of the field within the dielectric for a given amount of free charge on the plates.
- If two or more capacitors are connected in parallel, the potential difference is the same across all capacitors. The equivalent capacitance of a parallel combination of capacitors is

$$C_{\text{eq}} = C_1 + C_2 + C_3 + \dots$$
- If two or more capacitors are connected in series, the charge is the same on all capacitors, and the equivalent capacitance of the series combination is given by

$$1/C_{eq} = 1/C_1 + 1/C_2 + 1/C_3 + \dots$$

- Experiments shows that the charging process of a capacitor exhibits the exponential behaviour therefore we can write its equation as

$$Q = Q_0(1 - e^{-t/RC}) \quad (9.8)$$

- where as The discharging process of the capacitor follows the following equation.

$$Q = Q_0 e^{-t/\tau} \quad (9.9)$$

- where the quantity $\tau = RC$ is called the time constant or the capacitive time constant of the circuit.
- Energy is stored in a charged capacitor because the charging process is equivalent to the transfer of charges from one conductor at a lower electric potential to another conductor at a higher potential. The energy stored in a capacitor of capacitance C with charge Q and potential difference V is

$$U = \frac{1}{2} QV = \frac{1}{2} CV^2$$



Capacitor

Charge storing Device

Capacitance

Parallel Plate Capacitor

$$C = \frac{\epsilon_0 A}{d}$$

$$C_o = \frac{\epsilon_0 \epsilon_r A}{d}$$

Types of Capacitors

Series Combination

$$C_{eq} = C_1 + C_2 + C_3 + \dots$$

Parallel Combination

$$1/C_{eq} = 1/C_1 + 1/C_2 + 1/C_3 + \dots$$

Charging of Capacitors

$$Q = Q_0(1 - e^{-t/RC})$$

Discharging of Capacitors

$$Q = Q_0 e^{-t/\tau}$$

Energy stored in a capacitors

$$U = \frac{1}{2} QV$$

$$U = \frac{1}{2} CV^2$$

EXERCISE

Section (A): Multiple Choice Questions (MCQs)

- The capacitance of a capacitor is NOT influenced by
 - Plate thickness
 - Plate area
 - Plate separation
 - Nature of the dielectric
- What is the value of capacitance of a capacitor which has a voltage of 4V and has 16C of charge?
 - 2F
 - 4F
 - 6F
 - 8F
- Capacitors are used in electric power supply system to:
 - Improve power factor
 - Reduce line current
 - Provide voltage stability
 - switching
- In a variable capacitor, capacitance can be varied by:
 - Turning the rotatable plates in or out
 - Changing the plates
 - Sliding the rotatable plates
 - Changing the material of plates
- Energy stored in the capacitor is:
 - $E = \frac{1}{4} CV$
 - $E = \frac{1}{2} CV^2$
 - $E = CV^2$
 - $E = \frac{1}{2} CV$
- The time constant of a series RC circuit consisting of $100\mu F$ capacitor in series with a 100Ω resistor is.
 - 0.1 s
 - 0.1 ms
 - 0.01 s
 - 0.01 ms
- The charging of a capacitor through a resistance follows
 - linear law
 - square law
 - exponential law
 - none of the above
- When the total charge in a capacitor is doubled, the energy stored
 - remains the same
 - is doubled
 - is halved
 - is quadrupled
- The capacitance C is charged through a resistor R. The time constant of the charging circuit is given by
 - C/R
 - $1/RC$
 - RC
 - R/C
- Capacitor blocks:
 - alternating current
 - direct current
 - both alternating and direct current
 - neither alternating nor direct current

Section (B): Structured Questions

CRQs

- State the factors on which the capacitance of a parallel plate capacitor depends.
- Explain what is meant by dielectric constant (relative permittivity). State two physical properties desirable in a material to be used as the dielectric in a capacitor.
- Derive expressions for the combined capacitance of two capacitors (a) connected in series, (b) connected in parallel.

- Derive an expression for the energy stored in a capacitor C when there is a potential difference V between the plates.
- A capacitor gets a charge of 50C when it is connected to a battery of emf 5 V . Calculate the capacity of the capacitor.

ERQs

- Calculate combined capacitance of capacitors in series and in parallel.
- Prove that energy stored in a capacitor is $E = \frac{1}{2} CV^2$
- Describe the factors affecting the capacitance of a parallel plate capacitor?

Numericals

- The capacitance of air-filled parallel-plate capacitor is 1.3 pF . If the separation of the plates is doubled and wax is inserted between them. The new capacitance is 2.6 pF . Find the dielectric constant of the wax. (Ans. 4.0)
- Three capacitors have capacitances 10 , 50 and $25\text{ }\mu\text{F}$ respectively as shown in figure. Calculate (i) charge on each when connected in parallel to a 250 V supply (ii) total capacitance and (iii) potential difference across each when connected in series. (Ans. (i) 2.5 mC , 12.5 mC , 6.75 mC (ii) $85\text{ }\mu\text{F}$ (iii) 156.25 V , 62.5 V , 31.25 V)
- Three capacitors are connected through a potential difference of 100 volts as shown in figure. Find the charges and potential difference across each capacitor. (Ans. $160\text{ }\mu\text{C}$, $60\text{ }\mu\text{C}$, $100\text{ }\mu\text{C}$, $V_1 = 80\text{ V}$, $V_2 = 20\text{ V}$)
- Capacitor is charged through a large non-reactive resistance by a battery of constant voltage V . For this arrangement, if the capacitor has a capacitance of $10\text{ }\mu\text{F}$ and the resistance is $1\text{ M}\Omega$, calculate the time taken for the capacitor to receive 90% of its final charge. (Ans. 23 s)
- What capacitance is required to store energy of 10 kWh at a potential difference of 1000 V ? (Ans. 72 F)
- A $2.0\text{ }\mu\text{F}$ capacitor and a $4.0\text{ }\mu\text{F}$ capacitor are connected in parallel across a 300 V potential difference. Calculate the total energy stored in the capacitors. (Ans. 0.27 J)
- A 12.0-V battery is connected to a capacitor, resulting in $54.0\text{ }\mu\text{C}$ of charge stored on the capacitor. How much energy is stored in the capacitor?
(Ans. $3.24 \times 10^{-4}\text{ J}$)



Figure for 9.9

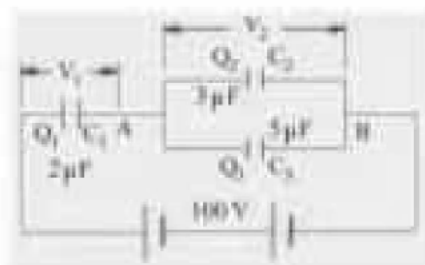


Figure for 9.10